

Malio OCR benchmark — honest ranking

Generated 2026-10-07T21:17:43+00:00 UTC · 4 adjudicated phone-photo invoices · 7 engine configs scored · 1 BLOCKED (missing license/SDK).

What this ranks: per-field exact-match correctness, using the shared regex extractor from pdf-tests/extractor.py. The engine with the most correct fields across the manifest wins. Ties are listed. Fields marked UNREADABLE or AMBIGUOUS in the ground truth are excluded from denominators.

n = 4. Wilson 95% CIs span ~45 percentage points. This ranking is informative, not statistically conclusive. Roadmap §6 asks for ~100 documents to screen engineering decisions. Confidence intervals narrow materially only above n ≈ 50.

Ranking — all scorable fields

Counts both PRESENT fields (engine must extract the correct value) and ABSENT fields (engine must return null). Scoring 3/4 on allocation number mostly reflects correctly returning null on 3 docs where no allocation exists — see the next table for a stricter view.

Rank	Engine	Correct / Scorable	Accuracy	Wilson 95% CI	Docs won
1	google-vision-document-text-detection-CLOUD-CONTROL	21 / 30	70.0%	[52.1%, 83.3%]	3.0
2	paddleocr-hebrew-server-svtrv2-line	15 / 30	50.0%	[33.2%, 66.8%]	0.3
3	kraken-medium-cpu	13 / 30	43.3%	[27.4%, 60.8%]	0.3
4	paddleocr-hebrew-server-svtrv2-word	12 / 30	40.0%	[24.6%, 57.7%]	0.3
5	ironocr-heb-ara-eng	11 / 30	36.7%	[21.9%, 54.5%]	0.0
6	tesseract-best-heb-eng-psm3	10 / 30	33.3%	[19.2%, 51.2%]	0.0
	tesseract-fast-heb-eng-psm3	10 / 30	33.3%	[19.2%, 51.2%]	0.0
—	abbyy-fre12-heb-ara-eng	BLOCKED	n/a	—	—

Honest view — PRESENT fields only (did the engine actually extract?)

Removes the inflation from easy ABSENT matches. This is the harder question: when a field IS on the document, does the engine recover the correct value?

Rank	Engine	Correct / Scorable (PRESENT)	Accuracy	Wilson 95% CI
1	google-vision-document-text-detection-CLOUD-CONTROL	18 / 27	66.7%	[47.8%, 81.4%]
2	paddleocr-hebrew-server-svtrv2-line	12 / 27	44.4%	[27.6%, 62.7%]
3	kraken-medium-cpu	10 / 27	37.0%	[21.5%, 55.8%]
4	paddleocr-hebrew-server-svtrv2-word	9 / 27	33.3%	[18.6%, 52.2%]
5	ironocr-heb-ara-eng	8 / 27	29.6%	[15.9%, 48.5%]
6	tesseract-best-heb-eng-psm3	7 / 27	25.9%	[13.2%, 44.7%]
	tesseract-fast-heb-eng-psm3	7 / 27	25.9%	[13.2%, 44.7%]
—	abbyy-fre12-heb-ara-eng	BLOCKED	n/a	—

Per-field correctness matrix

One cell per (engine × field). Shows correct/scorable where scorable > 0, otherwise '—'. Fields excluded from this document's ground truth (UNREADABLE/AMBIGUOUS) do not count.

Engine	Vendor	Tax ID	Invoice#	Date	BeforeVAT	VAT	Total	Alloc#
google-vision-document-text-detection-CLOUD-CONTROL	1/3	2/4	3/4	3/3	3/4	3/4	3/4	3/4
paddleocr-hebrew-server-svtrv2-line	0/3	2/4	3/4	1/3	2/4	2/4	2/4	3/4
kraken-medium-cpu	1/3	4/4	3/4	1/3	0/4	0/4	1/4	3/4
paddleocr-hebrew-server-svtrv2-word	0/3	2/4	2/4	1/3	1/4	1/4	2/4	3/4
ironocr-heb-ara-eng	1/3	3/4	1/4	0/3	1/4	1/4	1/4	3/4
tesseract-best-heb-eng-psm3	1/3	3/4	0/4	0/3	1/4	1/4	1/4	3/4
tesseract-fast-heb-eng-psm3	1/3	3/4	0/4	0/3	1/4	1/4	1/4	3/4
abbyy-fre12-heb-ara-eng	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD

Per-document winner (critical-fields-correct)

Which engine extracted the most fields correctly on each document. Ties shared.

Document	Winning engine(s)
Handwritten	kraken-medium-cpu paddleocr-hebrew-server-svtrv2-line paddleocr-hebrew-server-svtrv2-word
photo_2_2	google-vision-document-text-detection-CLOUD-CONTROL
photo_3_1	google-vision-document-text-detection-CLOUD-CONTROL
photo_4	google-vision-document-text-detection-CLOUD-CONTROL

Methodology

Candidates (local — six per roadmap §4): RivoksLab/paddleocr-hebrew (word, line), Tesseract + tessdata_best, Tesseract + tessdata_fast, Kraken (medium multilingual), IronOCR (2026.10.2 + Hebrew/Arabic language packs), ABBYY FineReader Engine 12.

Reference control: Google Cloud Vision DOCUMENT_TEXT_DETECTION, included as a CLOUD control (not one of the six candidates — roadmap §1 forbids hosted OCR in the main benchmark). Rendered in the same tables for direct comparison, labeled with -CLOUD-CONTROL suffix.

Scoring rules (roadmap §7): exact-match per field after NFC + whitespace normalization. IDs are strings (leading zeros preserved); amounts compared via Python Decimal at annotated precision; dates compared as ISO yyyy-mm-dd. No edit-distance tolerance in primary scoring.

Downstream parser: all engines pipe through the same deterministic regex extractor (pdf-tests/extractor.py). The parser is the bottleneck for every engine — its label patterns were built for text-layer PDFs, so Hebrew OCR noise (n↔n, n↔o substitutions, dot-vs-comma decimals) breaks label matching even when the raw OCR has the right tokens.

Ground truth: adjudicated by Claude reading each image directly (2026-10-07). Vendor name convention follows roadmap §2 (= the דוֹבָר/counterparty on the invoice). The handwritten receipt's invoice_date and the Migdal invoice's truncated vendor_name are marked AMBIGUOUS and excluded from their denominators.

What this ranking does NOT measure

- Latency, RAM, GPU, cost per 1M documents, cost per correct invoice — all roadmap §9 columns. Would require workload measurements + dated license quotes.
- Raw-text OCR quality — e.g., Google Vision captures amounts in raw text that the parser then fails to extract. Judged on the final canonical JSON, not the OCR substrate.
- Ensemble/routing performance — roadmap §5 options A/B/C routers are not yet simulated.
- Script-specific accuracy — e.g., Arabic-only subsets. n=4 is too small.
- Statistical ranking — Wilson CIs are too wide at n=4 to establish "engine X is better than engine Y" with confidence.